Supplemental Material for “Seed geometry in nucleation-controlled transition times in time-dependent Ginzburg–Landau models”

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SUPPLEMENTARY FIGURES

ORDERED-LIKE SEED STATISTICS

Figure S1 shows the statistics of ordered-like seeds, defined by

$$|\phi(\mathbf{x}, 0)| > 0.5\phi_s. \tag{1}$$

These quantities remain close to zero for all initial-state labels, indicating that the initial state does not typically contain pre-existing stable-phase droplets.

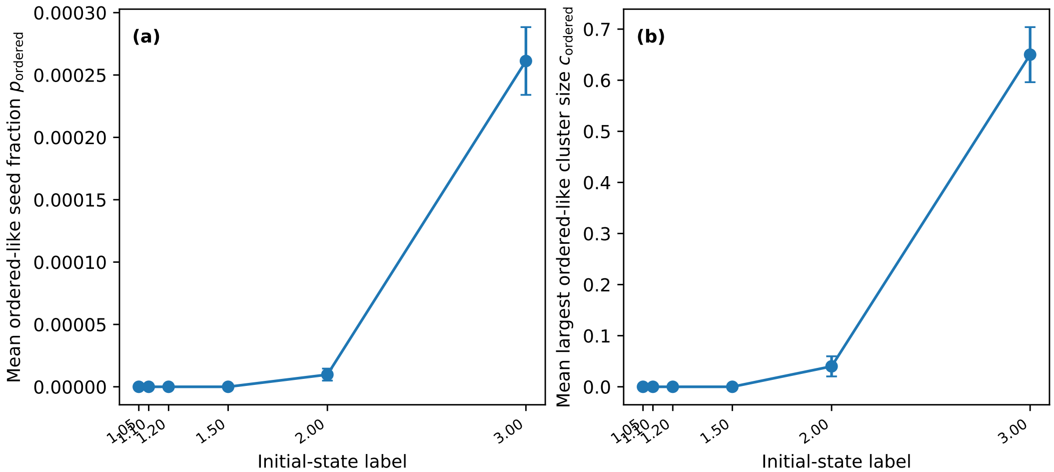

FIG. S1. Ordered-like seed statistics in the initial state. (a) Mean ordered-like seed fraction, defined by $|\phi(\mathbf{x}, 0)| > 0.5\phi_s$. (b) Mean largest ordered-like seed cluster size. Error bars indicate standard errors over stochastic realizations.

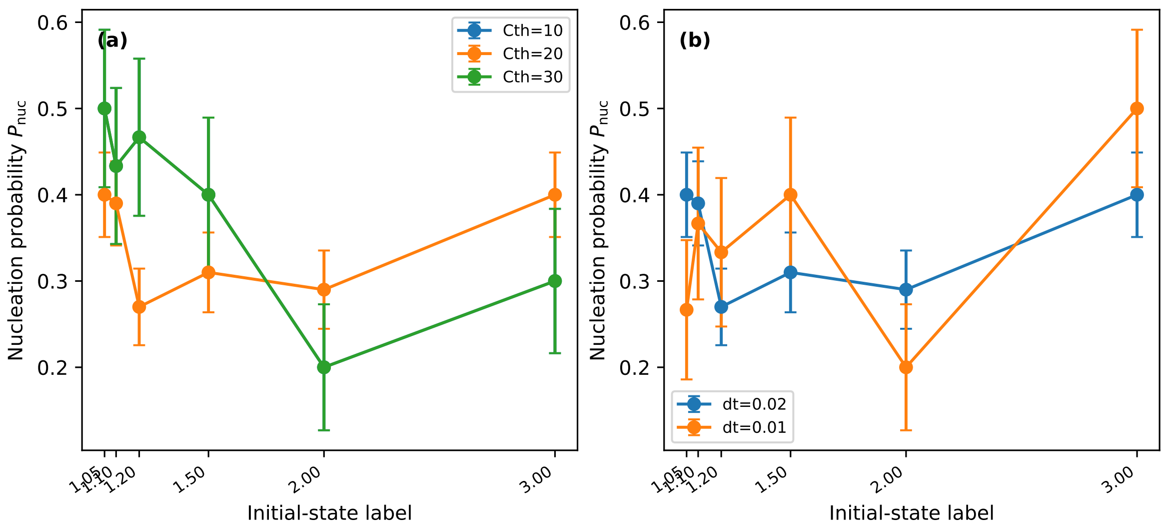

FIG. S2. Numerical robustness checks for the nucleation probability. (a) Dependence on the cluster threshold C_{th} used to identify nucleation events. (b) Dependence on the time step. Error bars indicate standard errors over stochastic realizations.

NUMERICAL ROBUSTNESS CHECKS

Figures S2 and S3 show numerical and operational robustness checks. The purpose of these checks is not to establish full finite-size convergence, but to verify that the central seed-quality trend is not an artifact of one particular numerical setting or nucleation criterion.

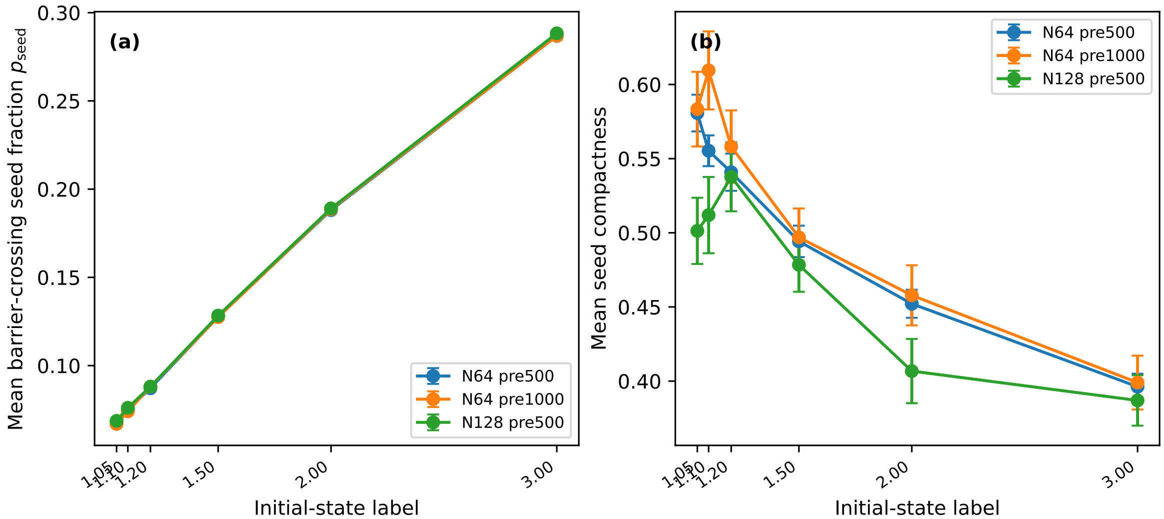

FIG. S3. Robustness of seed statistics against system size and pre-equilibration length. (a) Barrier-crossing seed fraction p_{seed}. (b) Seed compactness. The increase of p_{seed} and the decrease of seed compactness with the initial-state label are preserved in these checks. Error bars indicate standard errors over stochastic realizations.

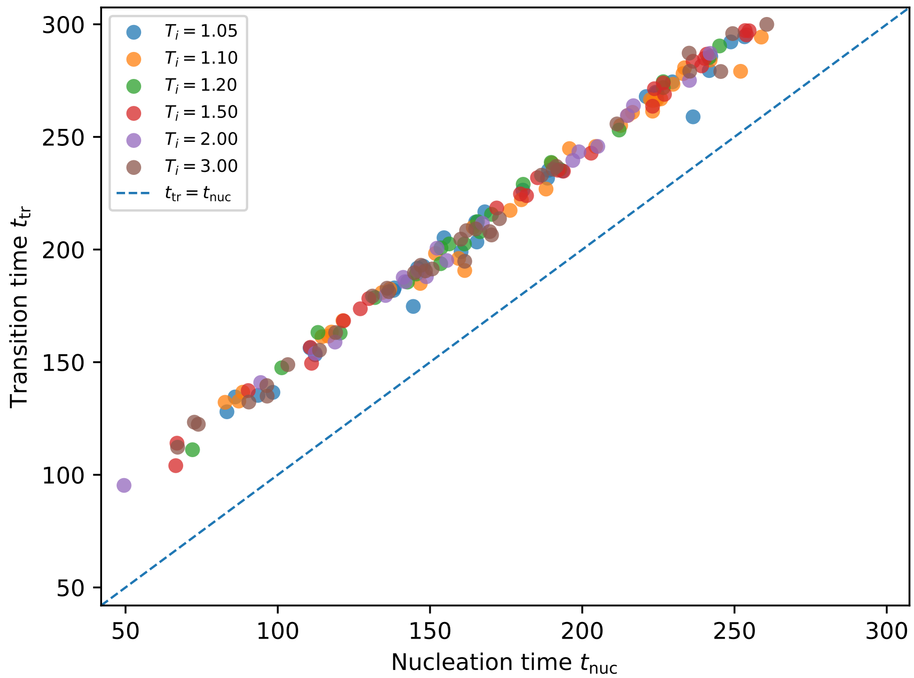

FIG. S4. Relation between the cluster nucleation time t_{nuc} and the global transition time t_{tr} in the first-order ϕ^6 model. Only samples for which both times were observed are shown. The dashed line indicates $t_{\mathrm{tr}} = t_{\mathrm{nuc}}$.

RELATION BETWEEN NUCLEATION AND TRANSITION TIMES

Figure S4 shows the relation between the cluster nucleation time t_{nuc} and the global transition time t_{tr} for samples in which both times were observed. The two times are strongly correlated. This supports the interpretation that the global transition time is closely tied to the stochastic nucleation time in the first-order model.

For the main first-order data set, the combined sample contains 600 realizations. Among these, 206 realizations show an observed cluster nucleation event and 165 realizations show an observed global transition by the final observation time. The scatter plot includes the 165 realizations for which both t_{nuc} and t_{tr} were observed. The Pearson correlation between these two times is approximately 0.997, and the Spearman rank correlation is approximately 0.995. The mean delay $t_{\mathrm{tr}} - t_{\mathrm{nuc}}$ is approximately 43.5 time units.

TABLE S1. Logistic-model comparison for the binary outcome of whether cluster nucleation was observed by the final observation time. Lower AIC is better. The sample contains 600 realizations, of which 206 nucleated by the final observation time.

Model	Predictors	ΔAIC
Seed quality	C_{comp}, R_g	0.00
Intercept only	none	1.08
Label only	T_i	2.87
Amount plus quality	$p_{\mathrm{seed}}, c_{\mathrm{seed}}, C_{\mathrm{comp}}, R_g$	3.70
Seed amount	$p_{\mathrm{seed}}, c_{\mathrm{seed}}$	4.09
All predictors	$T_i, p_{\mathrm{seed}}, c_{\mathrm{seed}}, C_{\mathrm{comp}}, R_g$	5.47

SAMPLE-LEVEL SEED-GEOMETRY ANALYSIS

As an additional check, we examined whether initial seed variables carry sample-level information about whether a cluster nucleation event is observed within the finite observation window. For each realization in the main first-order data set, we defined a binary outcome equal to one if t_{nuc} was observed by the final observation time and zero otherwise. We then fitted logistic models using standardized predictors. This analysis should not be interpreted as a committor calculation: it uses only a small set of initial geometric descriptors and a finite final observation time.

Table S1 summarizes the Akaike information criterion (AIC) values for several low-dimensional models of the observed finite-time nucleation outcome. The model using seed compactness and radius of gyration has the lowest AIC among the tested models, but its improvement over the intercept-only model is modest, with $\Delta\mathrm{AIC} = 1.08$. The combined model containing both seed-amount and seed-geometry variables is not preferred by AIC. Thus, the sample-level predictive evidence is modest and should be used only as a consistency check.

Table S2 shows the standardized coefficients for the model containing both seed-amount and seed-geometry variables. Seed compactness has a positive standardized coefficient, $\beta = 0.25$, with an approximate two-sided p value of 0.052. The coefficients of p_{seed}, c_{seed}, and R_g are not statistically decisive in this finite data set. This result is consistent with the physical interpretation that compactness may be relevant, but it is not by itself a quantitative

TABLE S2. Standardized coefficients in the logistic model containing both seed-amount and seed-geometry variables. Approximate standard errors and p values are based on the local Hessian of the log-likelihood.

Predictor	Standardized coefficient	Approx. p value	Odds ratio per 1 s.d.
p_{seed}	-0.090	0.618	0.914
c_{seed}	0.122	0.662	1.130
C_{comp}	0.250	0.052	1.284
R_g	0.168	0.568	1.182

nucleation-rate theory and should not be read as a committor estimate.

Figure S5 visualizes the same conservative sample-level check. The binned probabilities and within-p_{seed} comparisons are intended only as descriptive diagnostics. They do not replace a committor analysis or a direct critical-droplet calculation.

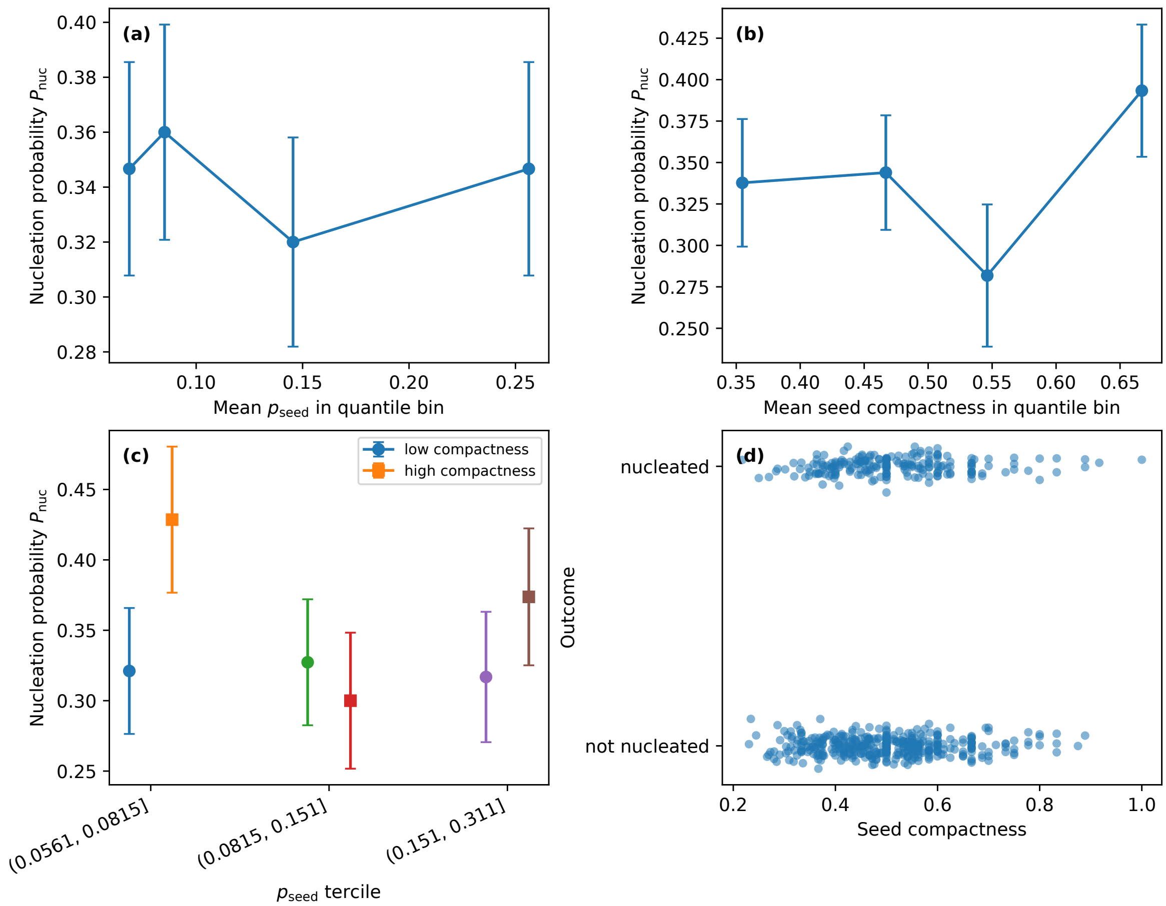

FIG. S5. Sample-level seed-quality diagnostics for the main first-order data set. (a) Nucleation probability binned by p_{seed}. (b) Nucleation probability binned by seed compactness. (c) Low- and high-compactness subsets compared within p_{seed} terciles. (d) Jittered binary nucleation outcome against compactness. These diagnostics provide only modest predictive support for compactness and are used as a conservative consistency check rather than as a committor analysis.